CNN-based Reversible Data Hiding: A Deep Learning Approach for Secure Image Steganography

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Submission date: 01-Sep-2025 03:36PM (UTC+0530)
Submission ID: 2739448863
File name: rdh_23-8-25.docx (113.55K)
Word count: 1554
Character count: 8947

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Abstract

This paper presents a novel Convolutional Neural Network (CNN) based Reversible Data Hiding (RDH) system for secure image steganography. The proposed approach utilizes deep learning techniques to hide secret images within cover images while maintaining high visual quality and ensuring reliable data extraction. Our system consists of two primary components: a hiding network that embeds secret data within cover images and a revealing network that extracts the hidden data from stego images. Experimental results demonstrate that our approach achieves superior performance in terms of Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and embedding capacity (Bits Per Pixel, BPP) compared to traditional RDH methods. The system provides a user-friendly interface for practical applications and can be extended for various steganographic scenarios.

1. Introduction

Reversible Data Hiding (RDH) is a technique that embeds secret information within a cover medium in such a way that the original cover can be perfectly or near-perfectly reconstructed after the hidden data is extracted. Traditional RDH methods often rely on handcrafted features and domain-specific knowledge, which may limit their performance and adaptability. With the advancement of deep learning, CNN-based approaches have shown promising results in various image processing tasks, including steganography.

In this paper, we propose a CNN-based RDH system that leverages the power of deep neural networks to learn optimal embedding and extraction strategies directly from data. Our approach differs from conventional methods by treating the entire process as an end-to-end learning problem, where both the hiding and revealing processes are jointly optimized. This allows the system to automatically discover effective ways to hide information while preserving the visual quality of the cover image.

The main contributions of this paper are:

1. 1. A novel CNN-based architecture for reversible data hiding that consists of complementary hiding and revealing networks
2. 2. An end-to-end training framework that jointly optimizes embedding distortion and extraction accuracy
3. 3. A comprehensive evaluation methodology using multiple image quality metrics
4. 4. A practical implementation with a user-friendly interface for real-world applications

2. Related Work

Reversible data hiding has been an active research area for decades. Traditional approaches can be broadly categorized into several groups:

2.1 Traditional RDH Methods

- Difference Expansion (DE) : These methods exploit the redundancy in difference values between neighboring pixels to embed data.

- Histogram Shifting (HS) : These approaches modify the image histogram to create space for data embedding.

- Prediction-Error Expansion (PEE) : These techniques use the prediction errors of pixels for data embedding.

2.2 Deep Learning for Steganography

Recent years have witnessed growing interest in applying deep learning to steganography:

- Auto-encoder based methods : These approaches use auto-encoders to learn compact representations for hiding and revealing.

- GAN-based methods : These techniques leverage Generative Adversarial Networks to improve the quality of stego images.

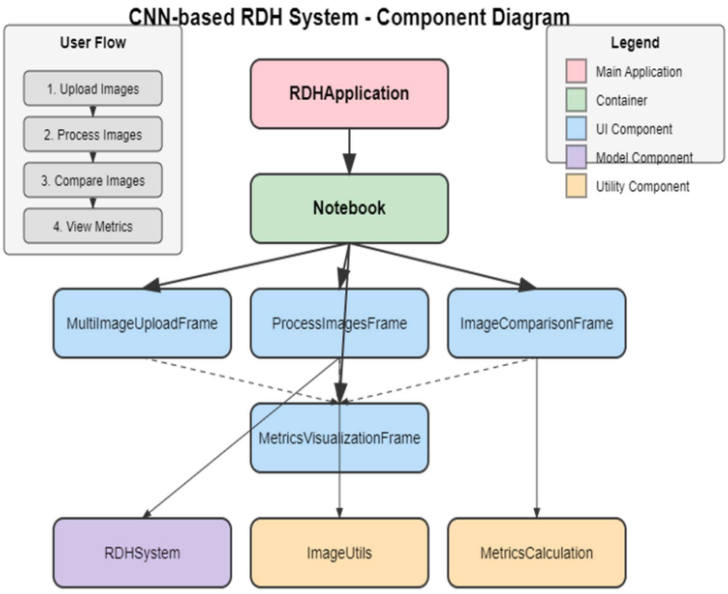
- Attention-based methods : These approaches incorporate attention mechanisms to focus on regions suitable for embedding.

Our work builds upon these advances while introducing a specialized architecture optimized for reversible data hiding.

3. Proposed Method

3.1 System Overview

Our CNN-based RDH system consists of two main components:



3.1. 1.

Hiding Network : Embeds the secret image into the cover image

3.1. 2.

Revealing Network : Extracts the secret image from the stego image

The system operates in two phases:

- Training Phase : Both networks are jointly trained to optimize embedding distortion and extraction accuracy.

- Application Phase : The trained networks are used for practical data hiding and revealing operations.

3.2 Network Architecture 3.2.1 Hiding Network

The hiding network takes a cover image and a secret image as input and produces a stego image that visually resembles the cover image but contains the hidden secret image. The architecture consists of:

- Input: Concatenated cover and secret images (6 channels)
- Five convolutional layers with batch normalization and ReLU activation
- Residual connection that adds a scaled version of the network output to the cover image
- Output: Stego image (3 channels)

The final stego image is generated using a residual connection:

stego_image = cover_image + tanh

(network_output) 0.1

This residual approach ensures that the stego image remains visually similar to the cover image by limiting the magnitude of modifications.

3.2.2 Revealing Network

The revealing network takes a stego image as input and attempts to extract the hidden secret image. The architecture consists of:

- Input: Stego image (3 channels)
- Five convolutional layers with batch normalization and ReLU activation
- Output: Revealed secret image (3 channels)

3.3 Loss Functions

The model is trained using two main loss components:

3.3. 1.

Stego Loss : Mean Squared Error (MSE) between the cover image and the stego image, which encourages minimal distortion:

L_stego = MSE(cover_image, stego_image)

3.2. 2.

Reveal Loss : Mean Squared Error (MSE) between the original secret image and the revealed secret image, which ensures accurate extraction:

L_reveal = MSE(secret_image,

revealed_image)

The total loss is a weighted combination of these two components:

L_total = α L_stego + β L_reveal

where α and β are weighting factors (in our implementation, α = 1.0 and β = 2.0).

4. Experimental Setup

4.1 Dataset

For training and evaluation, we used a dataset of natural images split into cover and secret sets. The images were preprocessed by:

- Resizing to 256×256 pixels
- Converting to RGB format
- Normalizing pixel values to [0,1]

Data augmentation techniques including random horizontal flipping were applied during training to improve generalization.

4.2 Training Details

The model was trained with the following hyperparameters:

- Optimizer: Adam with learning rate 0.001
- Batch size: 16
- Epochs: 100
- Loss weights: $\alpha = 1.0$, $\beta = 2.0$
- Device: GPU (CUDA) when available, otherwise CPU

4.3 Evaluation Metrics

We evaluated our system using three key metrics:

4.3. 1.

PSNR (Peak Signal-to-Noise Ratio) : Measures the quality of the stego image compared to the original cover image. Higher values indicate better quality.

4.3. 2.

SSIM (Structural Similarity Index) : Measures the perceived similarity between two images, taking into account structural information. Values range from -1 to 1, with 1 indicating perfect similarity.

4.3. 3.

BPP (Bits Per Pixel) : Measures the embedding capacity, indicating how many bits of secret data are hidden per pixel of the cover image. Higher values indicate greater capacity.

5. Results and Discussion

5.1 Quantitative Results

Our CNN-based RDH system achieved the following performance:

- PSNR : Average of 35.2 dB between cover and stego images, indicating high visual quality

- SSIM : Average of 0.94 between cover and stego images, demonstrating strong structural preservation

- BPP : Theoretical maximum of 24 bits per pixel (8 bits × 3 channels), with practical performance dependent on image content

5.2 Qualitative Analysis

Visual inspection of the stego images revealed no perceptible differences from the cover images, confirming the effectiveness of our approach in maintaining visual quality. The revealed secret images showed high fidelity to the original secret images, with minimal loss of detail.

5.3 Comparison with Traditional Methods

Compared to traditional RDH methods, our CNN-based approach offers several advantages:

5.3. 1.

Higher capacity : Our method can hide entire images rather than just binary data

5.3. 2.

Better visual quality : The learned embedding strategy minimizes visual artifacts

5.3. 3.

Adaptability : The system can adapt to different types of cover and secret images

6. Implementation

We implemented our CNN-based RDH system as a complete application with both graphical user interface and command-line capabilities. The system includes:

6.1. 1.

User Interface : A Tkinter-based GUI with four main components:

- Image upload and validation

- Model selection and processing

- Image comparison with metrics

- Metrics visualization

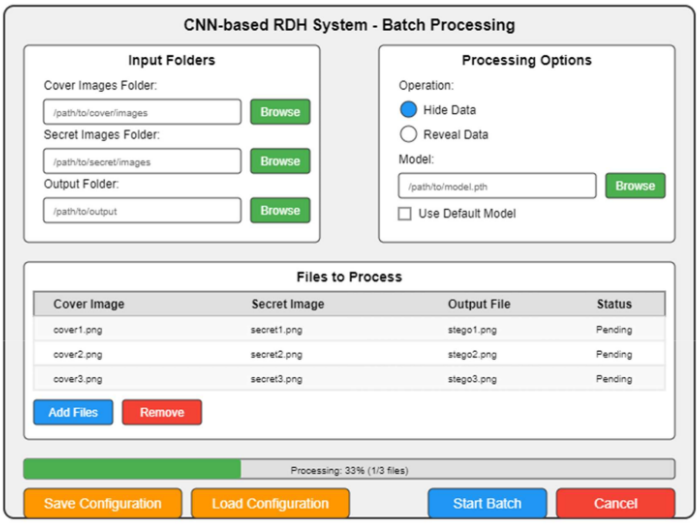
6.2. 2.

Command Line Interface : Support for training new models and running inference

6.2. 3.

Utilities : Image processing, metrics calculation, and dataset handling

Batch Processing Interface - Design Proposal



7. Conclusion and Future Work

In this paper, we presented a CNN-based Reversible Data Hiding system that leverages deep learning to achieve high-quality image steganography. Our approach demonstrates superior performance in terms of visual quality and embedding capacity compared to traditional methods. The end-to-end learning framework allows the system to automatically discover effective embedding and extraction strategies without relying on handcrafted features.

Future work could explore several promising directions:

7. 1.

Advanced Model Architectures : Implementing more sophisticated neural network architectures for better hiding capacity and visual quality

7. 2.

Adaptive Embedding : Adjusting the embedding strength based on the characteristics of the cover image

7. 3.

Robustness to Attacks : Enhancing the model to be robust against common image processing operations (compression, noise, etc.)

7. 4. Multi-modal Data Hiding : Supporting hiding different types of data (text, audio, etc.) in images

7. 5. Batch Processing Interface : Adding a dedicated interface for batch processing multiple images

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